

Subject:
Mother Languages (Greek)

Content:
(Vocabulary, Spelling, Grammar,
Oral skills, writing - recycling)

Computational practices:
choose what fits

- Algorithms
- Problem solving

Unplugged/plugged:
unplugged



Preparation time:
60min



Age:
6-10 years old



Duration:
60 - 80 min



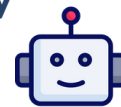
Location:
school, any classroom



Group distribution:
groups of 3-4 students



Difficulty
1/3



After this lesson, the children will be able to...

- separate materials and put them to the right recycling bin
- use correct vocabulary to describe the recyclable materials that can be put in each recycle bin colour
- use correctly the imperative mood
- create step-by-step movement plans for a robot
- identify movement patterns
- work together to solve challenges



Materials needed:

- Robot movement cards (move forward, turn left, turn right, stop, jump, spin, etc.) -see Annex
- Large floor grid (represents robot's world: home, school, city, playground) . You can use the grid in the Annex as an example
- Worksheets for writing captions in each recycling bin and giving information about recycling/reusing- see Annex.
Each group of students should have one worksheet.
- Stationery

Helpful links:

Information about recycling/reusing:

- <https://www.lifewithnogarbage.gr/>
- <https://inactionforabetterworld.com/en/in-action/>
- <https://wordwall.net/el/resource/26398760/%CE%B1%CE%BD%CE%B1%CE%BA%CF%85%CE%BA%CE%BB%CF%89%CF%83%CE%B7>

Acquiring basic robotic concepts about how to program a robot:

- <https://beebot.terrapiinlogo.com/>

Preparation and teacher notes

We prepare a small presentation about recycling /reusing.

Print and cut movement cards for each group. Each group should have a set of movement cards. A set consists of 10 move forward cards, 4 turn right, 4 turn left, a return-to-home card.

Create a large grid on the floor using tape or paper. You can also print the example grid found in the Annex. The floor grid represents the robot's environment and is divided into different sections, each representing a city block. In some blocks you can find a recycle bin and in others you see some garbage. Each garbage should be labeled clearly to help students link the kind of the garbage with the correct recycle bin.

We have to take into account that the recycling bin colours are different in different countries.

Warm up 20'

Introduce today's topic that is recycling.

We make a small presentation explaining:

- what recycling is and its importance
- the recyclable materials
- the colours of the recycle bins and which materials they refer to

We engage the children play an interactive game concerning the materials and the colours of the recycle bins to summarize what they have learnt through the presentation.

Build up 1/20'

After the game, we separate our students into groups of 3/4 students in each one. They are given a worksheet where they have to write a caption for each recycle bin and 2-3 phrases regarding the importance of recycling/reusing this type of material.

Examples:

In the red bins we can place used clothes or shoes. It is very important to place them there because some people that are in need may find them useful and avoid suffering from cold, etc.

Then, every group reads its work so as to summarize the captions and to hear everyone's thoughts regarding recycling/reusing. The teacher writes the caption on the board.

Preparation and teacher notes

Organize students into small groups (3-4 students).

Encourage teamwork and clear communication.

Encourage students to correctly use the imperative mood.

Build up 2/ 30'

The teacher shows the grid to the students. The main character is John and the large floor grid that represents the robot's environment, divided into squares where we can see garbage of different kind.

Students work into small groups (3-4 students).

Each group of students is provided with one grid and a set of movement cards. The teacher explains what each movement card does and shows examples. For example, the move forward card means that the robot will move one square on the grid.

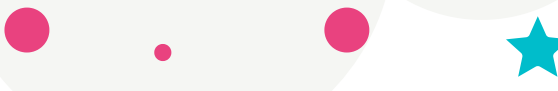
In addition to this, every student takes a card from a different deck that shows a unique item/garbage of the grid. Thus, in his turn, he has to program John collect this item and, then, throw it in the right bin. Then this exact item can be removed from the grid. Each group "programs" John to move around the grid to collect and throw the garbage step by step, using the movement cards to make John city clean. Students take turns guiding John through the grid.

Rehearsal 10'

When all teams are ready, they present their robot's route to clearing the city to the whole class. Students discuss the most effective and quickest routes to collecting the garbage.

While the robot moves, students must narrate its actions reminding to the class their colour choice of throwing the garbage.

If the robot makes a wrong choice, the class provides feedback and the team must troubleshoot by adjusting their instructions. The class gives also feedback regarding the correct use of the imperative mood. The class identifies movement patterns and suggests improvements to simplify the instructions and make the algorithm shorter.



Preparation and teacher notes

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Clean up 10'

The teacher gathers and organizes all materials and discusses key takeaways with students:

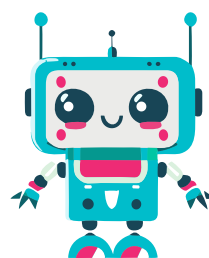
- "What was easy about programming your robot's routine?"
 - "What was challenging?"
 - "How did you fix mistakes in your robot's journey?"
 - "How can you improve your instructions (algorithm)?"
- 

John learns about recycling: Worksheet

This is John. John recently heard about recycling and has a bag full of recyclable materials, but is not sure about which recycling bin to use. Can you help him?

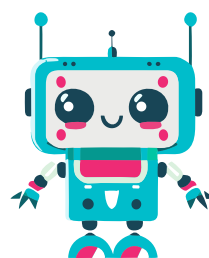
Put a caption for each recycle bin and write 2-3 sentences about how the materials that you mention for each recycle bin could be reused or what's the importance of their recycling

Hi! I'm John! I need your help in order to recycle!



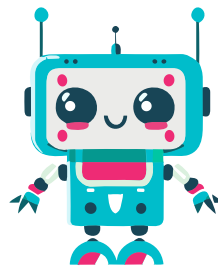
John learns about recycling: Worksheet

Hi! I'm John! I
need your help
in order to
recycle!



John learns about recycling: Worksheet

Hi! This is my city!



START



Annex: Robot movement cards

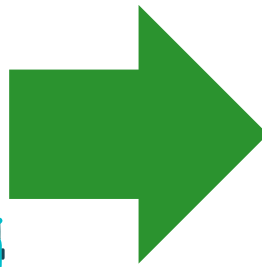
Robot movement cards

move forward



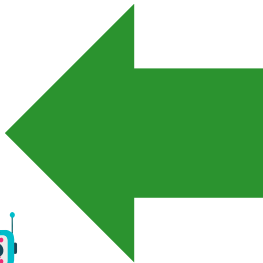
Robot movement cards

turn right



Robot movement cards

turn left



Robot movement cards

back to start



START

